Sovereign DevOps-Driven API Governance

Open Source and Agent-Ready in Hours

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API Peak

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Founder of **API Peak**: API strategy and governance services, community products

3scale	API Solution Engineer: API management, gateways, developer portals
adidas	API Evangelist: guidelines, API contract repository, CI quality gates
ING	Lead of Policy as Code in API governance
PZU	Chief API Architect: governance program from zero, 200+ external APIs, 160+ products, APIs ready for AI agents

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THE BOOK

RESTful API Design Patterns and Best Practices

Andrzej Jarzyna & Samir Amzani · Packt

Get it on Amazon

PARADIGM SHIFT

Your next API consumer is an agent

It takes your contract as is - completeness and quality are key

PZU · INTERNAL TEST OF AI AGENTS ON OUR APIS

Governed contracts make agents accurate

Single API calls

70–80% → **99.6%**

Bare OpenAPI → plus what governance already requires: descriptions, adherence to standards, examples

Multi-step workflows

~60% → **>90%**

Chained calls → plus Arazzo workflow descriptions on top of the same specs, or Hypermedia controls

AI-ready APIs are not a separate initiative. They are a product of your API governance program

WHERE IT BREAKS TODAY

An agent won’t ask. It’ll just call

1	It can’t find your API	**Catalog**
2	The contract is vague, and every team writes it differently	**Guidelines as code**
3	The contract promises one thing, the code does another	**Mocks & contract tests**
4	What’s running in production isn’t what the contract says	**Gateway from the contract**
5	One innocent change breaks it, and nobody notices	**Breaking-change check**

Governance is the developer experience of creating and using APIs. And the developer might now be an agent

Make the right thing easy, and the wrong thing hard

Premises

How we work

- Design-first
- The API contract (OpenAPI) is the single source of truth
- Configuration as code
- Runs on any CI. The demo uses Gitea Actions

Open source & sovereign

- No lock-in. Every tool is replaceable
- Small tools that do one job well, not one platform you customize for months
- Your data stays where you choose, under the law you choose
- No backdoors, no telemetry, no one you didn’t invite

In the demo: Backstage · Spectral · oasdiff · Microcks · KrakenD · Gitea. Swap any tool, keep the gates

THE STORY

One API, one partner

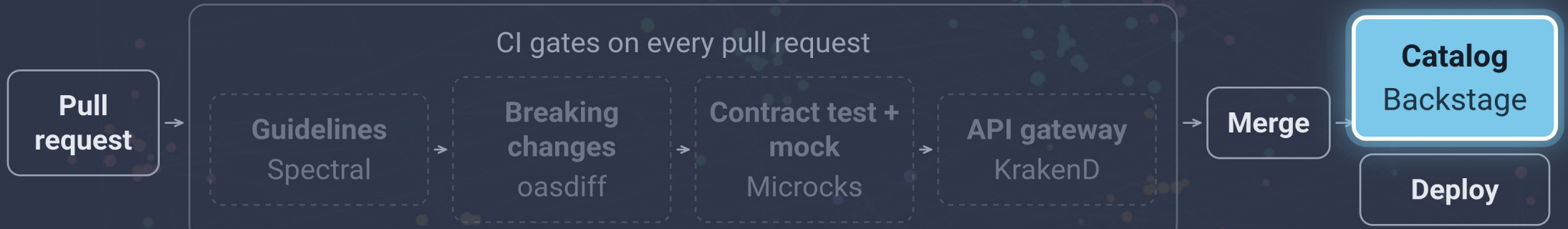
The **Orders API**: GET /orders/{orderId}. A fintech partner is building an AI agent on it

Each step adds one gate to this pipeline

STEP 1 — API CATALOG

You can’t govern what you can’t see

Thousands of APIs across the organization. Who knows they exist?



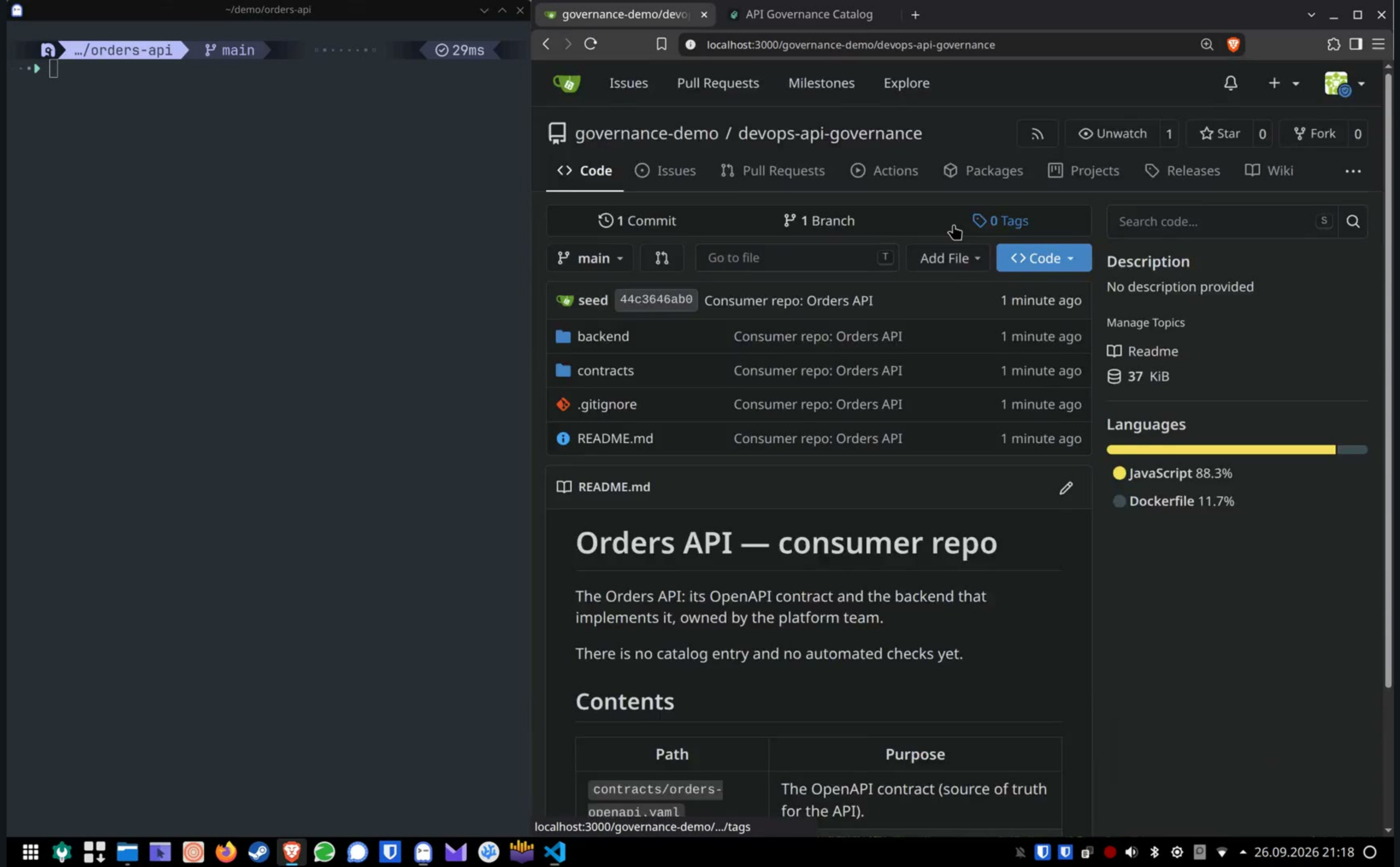
One file. That's the whole cost

```yaml
# catalog-info.yaml, next to the code
apiVersion: backstage.io/v1alpha1
kind: API
metadata:
  name: orders-api
spec:
  type: openapi
  lifecycle: experimental
  owner: group:default/platform-team
  definition:
    $text: ./contracts/orders-openapi.yaml
```

Name, owner, contract: next to the code

Merge it, and the API is discovered, visible org-wide, with its contract and owner in one place

The first step has to be almost invisible


~/demo/orders-api
…/orders-api
main
29ms
governance-demo/devo
API Governance Catalog
localhost:3000/governance-demo/devops-api-governance
Issues
Pull Requests
Milestones
Explore
governance-demo / devops-api-governance
Unwatch 1
Star 0
Fork 0
Code
Issues
Pull Requests
Actions
Packages
Projects
Releases
Wiki
1 Commit
1 Branch
0 Tags
Search code...
main
Go to file
Add File
Code
seed 44c3646ab0 Consumer repo: Orders API 1 minute ago
backend Consumer repo: Orders API 1 minute ago
contracts Consumer repo: Orders API 1 minute ago
.gitignore Consumer repo: Orders API 1 minute ago
README.md Consumer repo: Orders API 1 minute ago
Description
No description provided
Manage Topics
Readme
37 KiB
Languages
JavaScript 88.3%
Dockerfile 11.7%
README.md
Orders API — consumer repo
The Orders API: its OpenAPI contract and the backend that implements it, owned by the platform team.
There is no catalog entry and no automated checks yet.
Contents
Path
Purpose
contracts/orders-openapi.yaml
The OpenAPI contract (source of truth for the API).
localhost:3000/governance-demo/.../tags
26.09.2026 21:18

STEP 2 — GUIDELINES AS CODE

A guideline nobody enforces is a suggestion

At adidas, the real picture only showed up when the rules ran in CI

~/demo/orders-api

…/orders-api main 24ms

```
git switch feat/add-
```

governance-demo/devo | orders-api | API Governanc

localhost:3000/governance-demo/devops-api-governance

Issues Pull Requests Milestones Explore

governance-demo / devops-api-governance

Unwatch 1 Star 0 Fork 0

Code Issues Pull Requests Actions Packages Projects Releases Wiki

3 Commits 1 Branch 0 Tags

Search code... S

main Go to file T Add File Code

demo f6808d1aab	Merge pull request 'Register the Orde...	3 minutes ago
backend	Consumer repo: Orders API	5 minutes ago
contracts	Consumer repo: Orders API	5 minutes ago
.gitignore	Consumer repo: Orders API	5 minutes ago
catalog-info.yaml	Register the Orders API in the catalog	5 minutes ago
README.md	Register the Orders API in the catalog	5 minutes ago

README.md

Orders API — consumer repo

This is the **product/consumer repository**: the API contract, its catalog entry, and the backend implementation. It is the unit a product team owns and opens pull requests against.

Contents

Path	Purpose
`contracts/orders-openapi.yaml`	The OpenAPI contract (source of truth for the API).

Description

No description provided

Manage Topics

Readme

39 KiB

Languages

JavaScript 88.3%

Dockerfile 11.7%

26.09.2026 21:22

STEP 2 — TAKEAWAY

Feedback in seconds, not weeks

```
error  api-peak:rest17:2025-https-required
       server.url MUST use HTTPS
       → rule in the catalog: #https-api-peakrest172025-https
```

Rule, file, line and how to fix it, right on the pull request. The same rules for humans and CI: one source of truth

Spotlight

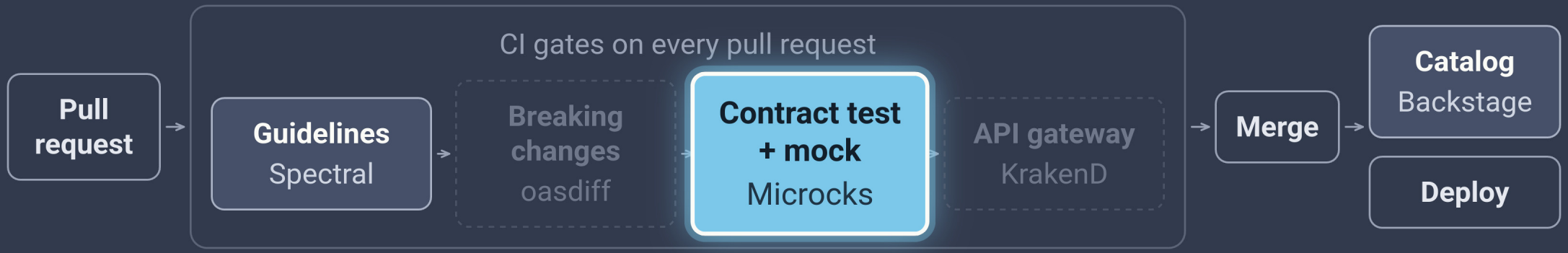
Spectral 6.16 sends telemetry. Spotlight fork fixes it (will rename to OpenLint)

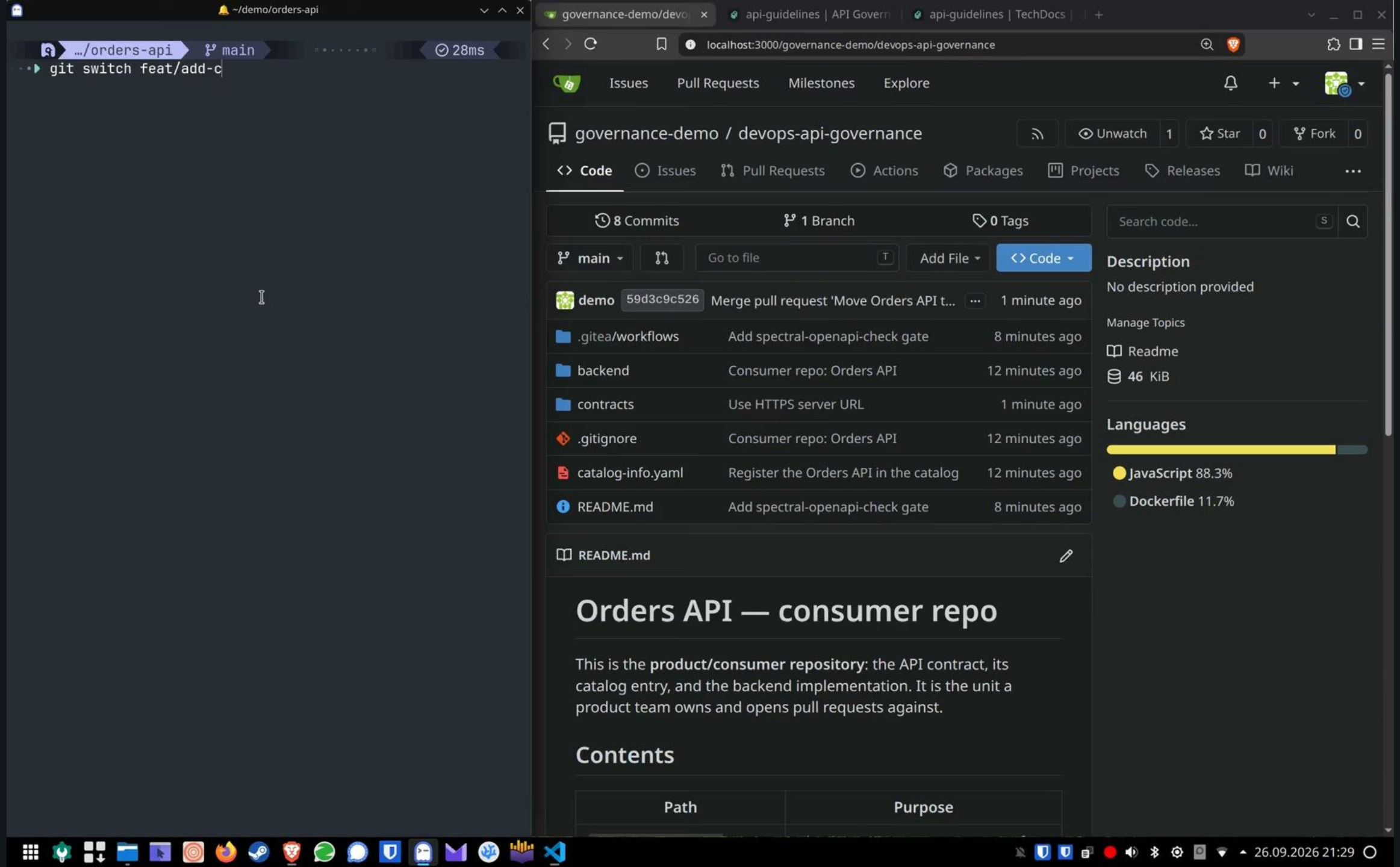
FOR AGENTS descriptions and examples in the contract are what lifted accuracy to 99.6%

STEP 3 — MOCKS AND CONTRACT TESTING

A contract is a promise

Can the partner integrate before the API exists? Does the code keep the promise?




~/demo/orders-api
…/orders-api
main
28ms
git switch feat/add-c
governance-demo/devo
api-guidelines | API Governa
api-guidelines | TechDocs
localhost:3000/governance-demo/devops-api-governance
Issues
Pull Requests
Milestones
Explore
governance-demo / devops-api-governance
Unwatch 1
Star 0
Fork 0
Code
Issues
Pull Requests
Actions
Packages
Projects
Releases
Wiki
8 Commits
1 Branch
0 Tags
Search code...
main
Go to file
Add File
Code
demo 59d3c9c526 Merge pull request 'Move Orders API t... 1 minute ago
.gitea/workflows Add spectral-openapi-check gate 8 minutes ago
backend Consumer repo: Orders API 12 minutes ago
contracts Use HTTPS server URL 1 minute ago
.gitignore Consumer repo: Orders API 12 minutes ago
catalog-info.yaml Register the Orders API in the catalog 12 minutes ago
README.md Add spectral-openapi-check gate 8 minutes ago
Description
No description provided
Manage Topics
Readme
46 KiB
Languages
JavaScript 88.3%
Dockerfile 11.7%
README.md
Orders API — consumer repo
This is the product/consumer repository: the API contract, its catalog entry, and the backend implementation. It is the unit a product team owns and opens pull requests against.
Contents
Path
Purpose
26.09.2026 21:29

STEP 3 — TAKEAWAY

One example, many uses

```
examples:
  order_123:
    value: { orderId: "123", isPaid: true }
# → the live mock AND the contract test
```

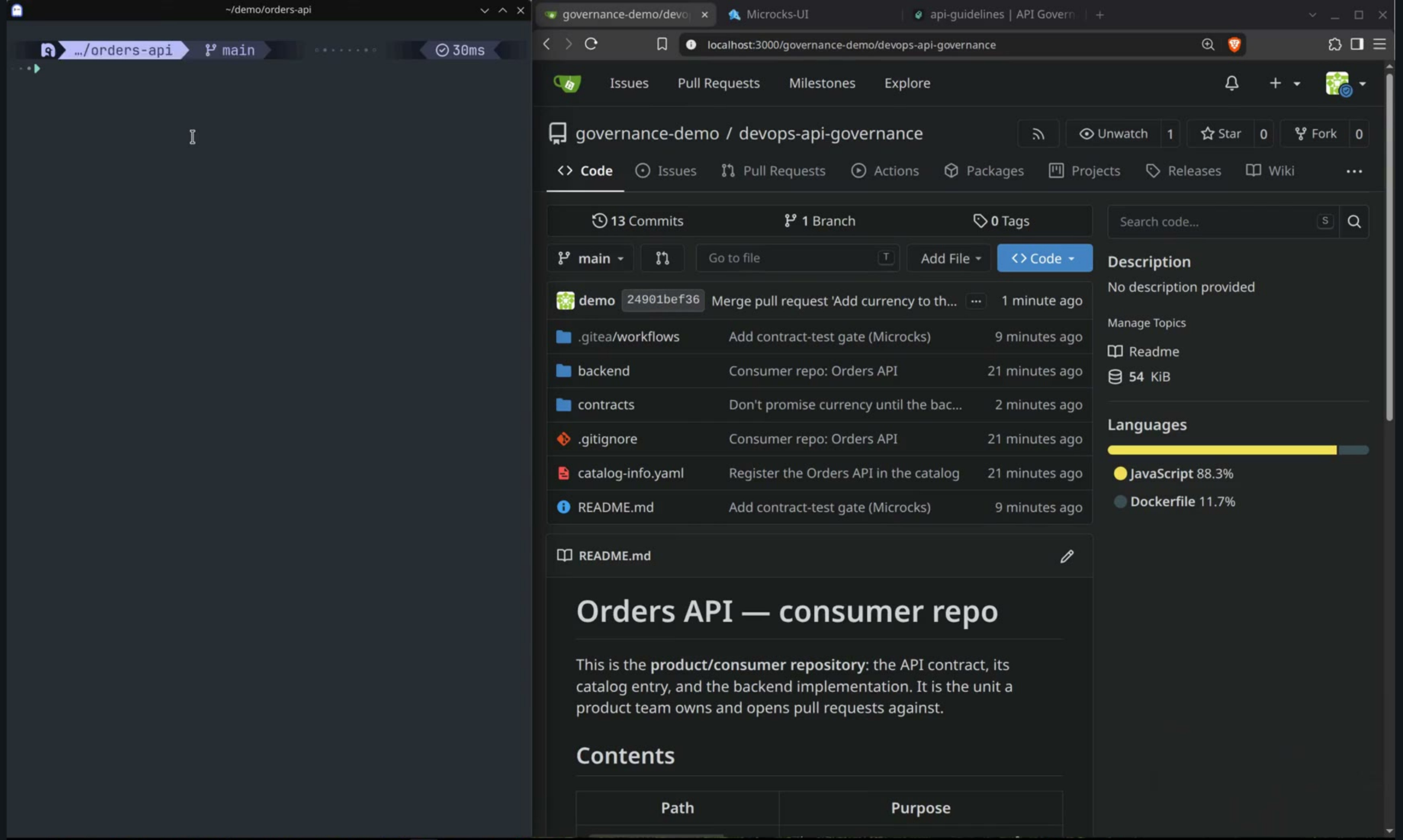
The partner builds against the mock before the code exists; CI proves the code keeps the promise on every pull request

FOR AGENTS build and test the agent against the mock before the API exists

STEP 4 — PUBLISH THROUGH THE API GATEWAY

Publish the API straight from its contract

Publishing is a lifecycle step, not a ticket. Hand-written gateway config drifts from the contract


~/demo/orders-api
…/orders-api
main
30ms
governance-demo/devo
Microcks-UI
api-guidelines | API Govern
localhost:3000/governance-demo/devops-api-governance
Issues
Pull Requests
Milestones
Explore
governance-demo / devops-api-governance
Unwatch 1
Star 0
Fork 0
Code
Issues
Pull Requests
Actions
Packages
Projects
Releases
Wiki
13 Commits
1 Branch
0 Tags
Search code...
main
Go to file
Add File
Code
demo 24901bef36 Merge pull request 'Add currency to th... 1 minute ago
.gitea/workflows Add contract-test gate (Microcks) 9 minutes ago
backend Consumer repo: Orders API 21 minutes ago
contracts Don't promise currency until the bac... 2 minutes ago
.gitignore Consumer repo: Orders API 21 minutes ago
catalog-info.yaml Register the Orders API in the catalog 21 minutes ago
README.md Add contract-test gate (Microcks) 9 minutes ago
Description
No description provided
Manage Topics
Readme
54 KiB
Languages
JavaScript 88.3%
Dockerfile 11.7%
README.md
Orders API — consumer repo
This is the product/consumer repository: the API contract, its catalog entry, and the backend implementation. It is the unit a product team owns and opens pull requests against.
Contents
Path
Purpose
26.09.2026 21:39

STEP 4 — TAKEAWAY

Generated, deployed, proven

```
$ curl -i localhost:8090/orders/123   # before
HTTP/1.1 404 Not Found
$ curl -i localhost:8090/orders/123   # after the merge
HTTP/1.1 200 OK  {"orderId":"123","isPaid":true}
```

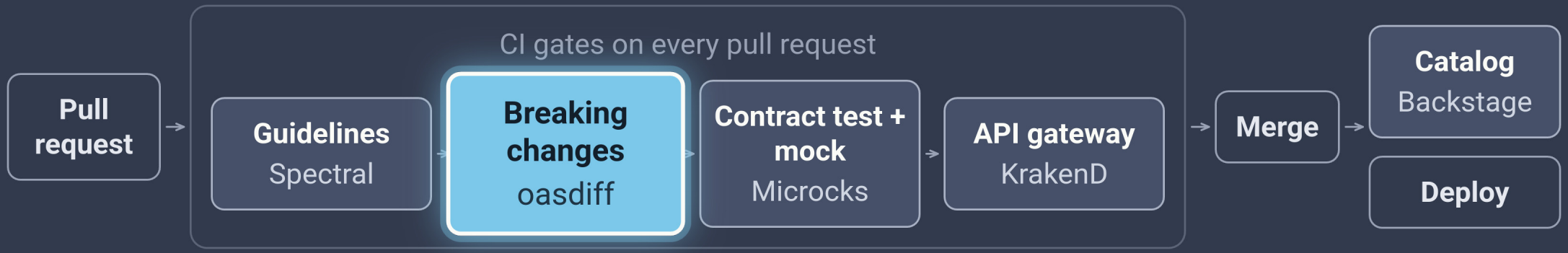
One endpoint per path and method in the contract, nothing else. The same contract test runs through the gateway: it must not change what the contract promises

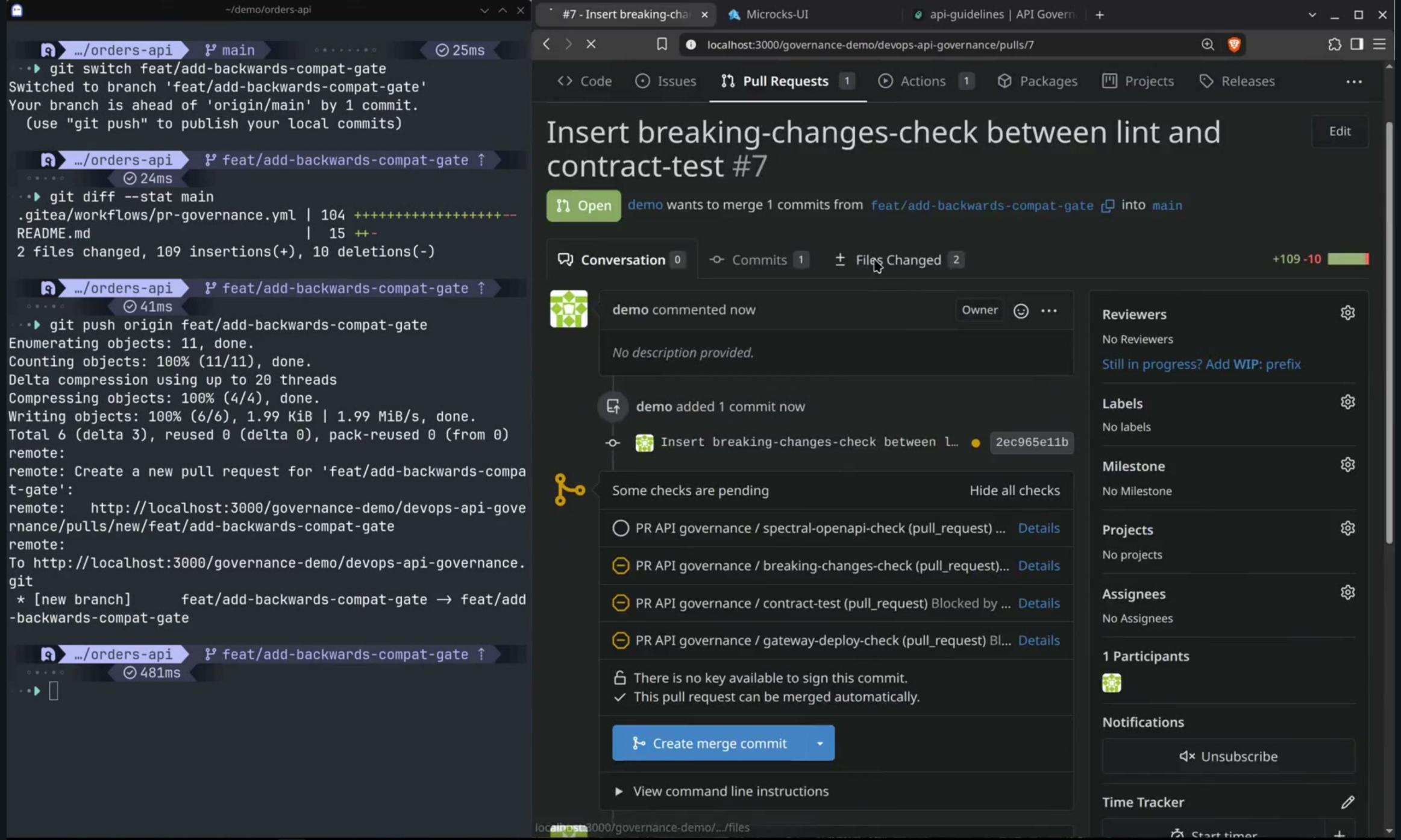
FOR AGENTS what runs in production is exactly what the agent read

STEP 5 — BREAKING CHANGES

Now the partner is integrated

A small, innocent change to the contract. Who finds out first: CI, or the partner?




~/demo/orders-api
…/orders-api main 25ms
git switch feat/add-backwards-compat-gate
Switched to branch 'feat/add-backwards-compat-gate'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
…/orders-api feat/add-backwards-compat-gate ↑
24ms
git diff --stat main
.gitea/workflows/pr-governance.yml | 104 ++++++++++++++++++--
README.md | 15 ++-
2 files changed, 109 insertions(+), 10 deletions(-)
…/orders-api feat/add-backwards-compat-gate ↑
41ms
git push origin feat/add-backwards-compat-gate
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 1.99 KiB | 1.99 MiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a new pull request for 'feat/add-backwards-compat-gate':
remote: http://localhost:3000/governance-demo/devops-api-governance/pulls/new/feat/add-backwards-compat-gate
remote:
To http://localhost:3000/governance-demo/devops-api-governance.git
* [new branch] feat/add-backwards-compat-gate → feat/add-backwards-compat-gate
…/orders-api feat/add-backwards-compat-gate ↑
481ms
#7 - Insert breaking-cha
Microcks-UI
api-guidelines | API Govern
localhost:3000/governance-demo/devops-api-governance/pulls/7
Code
Issues
Pull Requests 1
Actions 1
Packages
Projects
Releases
Insert breaking-changes-check between lint and contract-test #7
Edit
Open
demo wants to merge 1 commits from feat/add-backwards-compat-gate into main
Conversation 0
Commits 1
Files Changed 2
+109 -10
demo commented now
Owner
No description provided.
demo added 1 commit now
Insert breaking-changes-check between l…
2ec965e11b
Some checks are pending
Hide all checks
PR API governance / spectral-openapi-check (pull_request) … Details
PR API governance / breaking-changes-check (pull_request)… Details
PR API governance / contract-test (pull_request) Blocked by … Details
PR API governance / gateway-deploy-check (pull_request) Bl… Details
There is no key available to sign this commit.
This pull request can be merged automatically.
Create merge commit
View command line instructions
Reviewers
No Reviewers
Still in progress? Add WIP: prefix
Labels
No labels
Milestone
No Milestone
Projects
No projects
Assignees
No Assignees
1 Participants
Notifications
Unsubscribe
Time Tracker
Start timer
localhost:3000/governance-demo/…/files
26.09.2026 21:44

STEP 5 — TAKEAWAY

An explicit decision, not an accident

```
error [new-required-request-parameter]
  in API GET /orders/{orderId}
  added the new required `query` request parameter `channel`
exit 1 → PR blocked (contract test + gateway: skipped)
```

Blocked before merge, before deploy, before the incident. Make it optional, or ship a new major version on purpose (rule rest40)

FOR AGENTS a human reads the changelog. An agent just fails, at scale, often silently

THE WHOLE PIPELINE

You don’t need five gates on day one

Add one at a time, like we just did. Every change is a pull request with recorded gate results: an audit trail for free.
Open source, sovereign, one contract

Monday morning

1

Track all your APIs in one place, even from a monorepo

2

Turn your top 5 guideline rules into a CI lint, warnings first

3

Run a breaking-change diff on your most-used API

Your APIs get better for people, and ready for agents, as a by-product

Thank you!

Andrzej Jarzyna · API Peak

Run the whole demo yourself: **docker compose up**

The repo, on Codeberg

codeberg.org/pierogi/devops-api-governance

FOST LONDON 2026

How was this session?

Scan to rate it. It takes a minute and it really helps

Questions? I’m here now, and after the session